# Guide Creation Request

Here is a **clean, exam-ready step-by-step guide** based on your PDF (installation steps skipped as requested).

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## ✅ Jenkins Lab Guide (Step-by-Step)

📄 Source:

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## 🔹 PART 1: Jenkins Basics & First Job

### 1. Open Jenkins

- Launch Jenkins in browser (`http://localhost:8080`)
- Complete initial setup:
  - Unlock Jenkins (admin password)
  - Install suggested plugins
  - Create admin user
  - Finish configuration

*(Derived from setup screens shown on pages 6–8)*

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### 2. Understand Jenkins UI

From dashboard:

- **Dashboard** → shows all jobs
- **New Item** → create job
- **Manage Jenkins** → system + plugins
- **Build History / Queue** → running & past jobs

*(Pages 8–10)*

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### 3. Create Freestyle Job

Steps:

1. Click **New Item**
2. Enter job name
3. Select **Freestyle Project**
4. Click **OK**

*(Pages 10–11)*

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## 4. Configure Simple Build

1. Scroll to **Build Section**
2. Click **Add Build Step**
3. Select:
   - Windows → *Execute Windows batch command*
4. Add:

```
[redacted]
```

5. Click **Save**

*(Page 12)*

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## 5. Run Job Manually

- Click **Build Now**
- Open:
  - **Build History**
  - **Console Output**

✔ Output should show:

```
[redacted]
```

*(Page 13)*

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# 🔹 PART 2: Git Integration with Jenkins

## 6. Create GitHub Repository

- Create a new repository
- Copy repo URL

*(Page 15)*

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## 7. Configure Git in Jenkins

1. Go to **Manage Jenkins**
2. Open **Global Tool Configuration**
3. Under Git:
    - Click **Add Git**
    - Provide Git path/name
4. Save

*(Page 16)*

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## 8. Create Job with Git

1. Create **Freestyle Job**
2. Go to:
    - **Source Code Management → Git**
3. Paste repository URL
4. Save

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## 9. Add Build Step

- Add:
    - Windows batch / Shell command

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## 10. Enable Auto Build (Trigger)

- Go to **Build Triggers**
- Select:
    - **Poll SCM** OR
    - **Webhook**

✔️ Enables automatic build on code changes

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# 🔹 PART 3: Job Chaining (Build → Test → Deploy)

## 11. Create Multiple Jobs

Create:

- Job-A → Build
- Job-B → Test
- Job-C → Deploy

*(Pages 19–20)*

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## 12. Configure Job Chaining

### Step 1: Link Job-A → Job-B

1. Open **Job-A**
2. Click **Configure**
3. Scroll to **Post-build Actions**
4. Select:
    - **Build other projects**
5. Enter: `Job-B`

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### Step 2: Link Job-B → Job-C

- Repeat same steps
- Add `Job-C`

✔️ Ensure:

- Trigger only if build is stable

*(Pages 21–22)*

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### 13. Visualize Job Flow

- Run Job-A
- Observe:
  - Job-B triggers automatically
  - Job-C runs after Job-B

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## 🔹 PART 4: Pipeline & Stage View

### 14. Install Pipeline Stage View

1. Go to **Manage Jenkins**
2. Click **Manage Plugins**
3. Go to **Available**
4. Search:
   - `Pipeline Stage View`
5. Install (without restart)

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### 15. Create Pipeline Job

1. Click **New Item**
2. Select **Pipeline**
3. Click OK

*(Page 25)*

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### 16. Write Pipeline Script (UI)

Basic structure:

[redacted]

[redacted]

*(Pages 25–26)*

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### 17. Run Pipeline

- Click **Build Now**
- Observe:
  - Stage View (Build → Test → Deploy)

*(Page 27)*

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## 🔹 PART 5: Pipeline as Code (Jenkinsfile)

### 18. Create Jenkinsfile

- Create file in repo:

[redacted]

- Add pipeline code (same as above)

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### 19. Push to GitHub

- Commit & push Jenkinsfile

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## 20. Configure Pipeline Job (Git)

1. Open pipeline job
2. Select:
   - **Pipeline from SCM**
3. Choose:
   - Git
4. Enter repo URL

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## 21. Enable Auto Trigger

- Use:
  - Webhook / Poll SCM

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## 22. Modify & Rebuild

- Update Jenkinsfile
- Push changes
- Jenkins auto rebuilds

*(Page 31)*

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## 23. View Output

- Open:
  - **Console Output**
  - **Stage View**

✔ Shows stage-wise execution

*(Pages 32–33)*

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# ✅ FINAL FLOW SUMMARY

[redacted]

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If you want, I can convert this into:

- **Viva answers**
- **Short notes (2–3 marks)**
- **Flow diagram for exam writing**